Trần Tiến

Full-Stack & IoT Engineer

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OBJECTIVE

To leverage my expertise in embedded systems and modern software development to build seamless, high-performance IoT products. My immediate goal is to join an engineering team where I can contribute to both firmware optimization and scalable web/desktop application development. In the long term, I aim to become an **IoT Solution Architect**, leading the technical design and deployment of comprehensive, intelligent hardware-software ecosystems.

PROJECTS

Yukinu

01/2025 - 05/2026

Architected the Monorepo structure, decoupled core logic into internal packages, and enforced absolute end-to-end Type-safety using TypeScript.

Yukinu is a comprehensive multi-vendor e-commerce platform built on a Monorepo architecture to optimize performance and code reusability. The project features a customer-facing Storefront and an advanced Admin Dashboard for vendors. All core logic - including Authentication, API, Database, and UI Components - is decoupled into independent internal packages. This ensures seamless system synchronization and guarantees end-to-end Type-safety from the database to the user interface using TypeScript.

- Github: https://github.com/tiesen243/yukinu
- Preview: https://yukinu.vercel.app

Alarm Clock

04/2026 - 04/2026

Developed embedded firmware (C/C++) on Nios II for real-time hardware control and engineered a lightweight Tauri desktop app via UART.

Alarm Clock is an intelligent alarm system bridging FPGA hardware with a Tauri desktop controller. The hardware utilizes Verilog HDL and a Nios II soft-core processor to handle real-time clock management and hardware interrupts over the Avalon bus. The central control software is built using Tauri, communicating via UART for real-time synchronization and alarm scheduling with a minimal memory footprint.

- Github: https://github.com/tiesen243/alarm-clock

ERP

09/2025 - 12/2025

Implemented the RBAC user management system using PHP MVC and containerized the application with Docker to standardize the team's development environment.

A comprehensive enterprise web application designed to simulate and optimize the production value chain of a mechanical keyboard factory. The system automates complex workflows including order intake, production scheduling, inventory tracking, quality control, payroll management, and invoicing.

- Github: https://github.com/tiesen243/422000422702-QuanLySanXuat

CNN

04/2024 - 05/2024

Mathematically derived and coded the Backpropagation calculus for 2D Convolution and Max Pooling layers using optimized vectorized NumPy operations.

A pure NumPy implementation of a Convolutional Neural Network (CNN) trained on the MNIST handwritten digit dataset without high-level ML frameworks. The project features custom-built core layers including Convolutional (2D), Max Pooling, Flatten, and Fully Connected layers, achieving high classification accuracy through manual vectorization and mathematical optimization.

- Github: https://github.com/tiesen243/cnn-from-scratch

Egg Community

04/2024 - 04/2024

Engineered the relational database schema and optimized query performance for core social features including post CRUD, like interactions, and user follow mechanics.

Egg Community is a minimalist social networking platform inspired by Meta's Threads, built to optimize user engagement through core interactive features. The platform focuses on high performance and clean data relations, allowing users to seamlessly create, edit, delete text-based posts, react via likes, and manage social connections through a flexible follow/unfollow system.

- Github: https://github.com/tiesen243/egg-community

SKILLS

Programming Languages

C/C++, Python, Typescripts, Javascript

Software Development

Node.js, Next.js, React, React Native, Tailwind CSS, PostgreSQL

Embedded System

Verilog HDL, Nios II Soft-core, FPGA Design, Microcontrollers

Tools

Linux, Docker, Git, GitHub Actions

EDUCATIONS

(September 2022 - Present)

Computer Engineering Technology

Industrial University of Ho Chi Minh City

- **GPA:** 2.94/4.0

CERTIFICATES

07/2025

Basic Information Technology Applications Certificate